

Network Security - Vijeta Ma'am.

12/1/26.

★ IP Addressing.

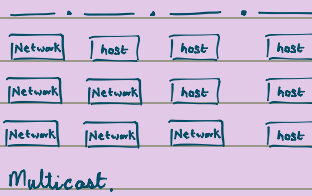
• Octet (8-bits)

• Class A

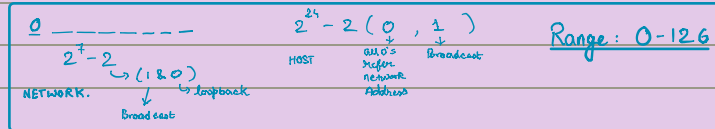
B

C

D



Class A:

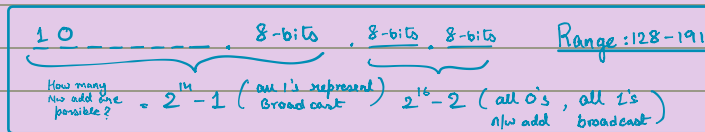


CIDR on Slash Notation.

N/w . 0's . 0's . 0's / 8

Subnet mask: 11111111.0.0.0 = 255.0.0.0
↓
used to Calculate Network Portion.

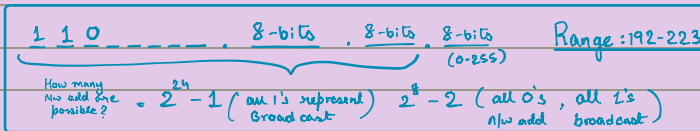
Class B:



N/w . N/w . 0's . 0's / 16

Subnet mask: 11111111.11111111.0.0 = 255.255.0.0
↓
used to Calculate Network Portion.

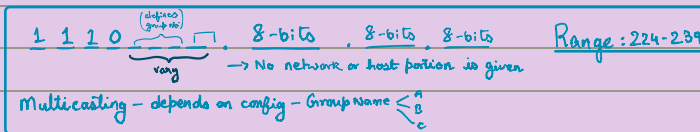
Class C:



N/w . N/w . N/w . 0's / 24

Subnet mask: 11111111.11111111.11111111.0 = 255.255.255.0
↓
used to Calculate Network Portion.

Class D:



Here Group No is referred as Subnet mask.

Q. If I need 80 groups then? 80 in binary needs 7 bits
How many machines can be added?

How many bits to network? - No. of Multicasting
Eg: 230.225.6.7 / 11
11100000.11100000.0's.0's
= 230.224.0.0 → Group No

29th Jan.

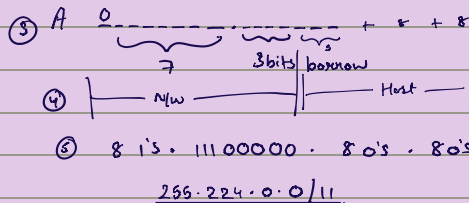
Subnetting

→ Larger network can be broken down into smaller networks by borrowing a bit from previous host.

→ Class A - fix bits are less.

260 = 9 bits
8 + 1

Eg: 560 networks requirement. ② In class A 7 bits, 8 14 bits.
① → 10 bits.



2 bits borrow
7 + 2 = 10

Q2

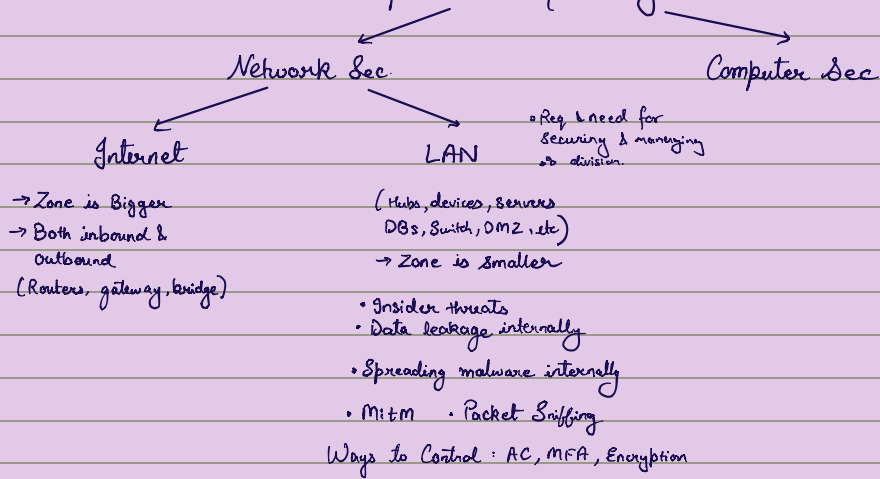
510
→ 9 bits → Class A → 7 (Borrow 2 bits)

8 1's . 11000000 + 8 0's . 80's

Ans. 255.192.0.0 / 10 = SM

→ always +1 than no of bits

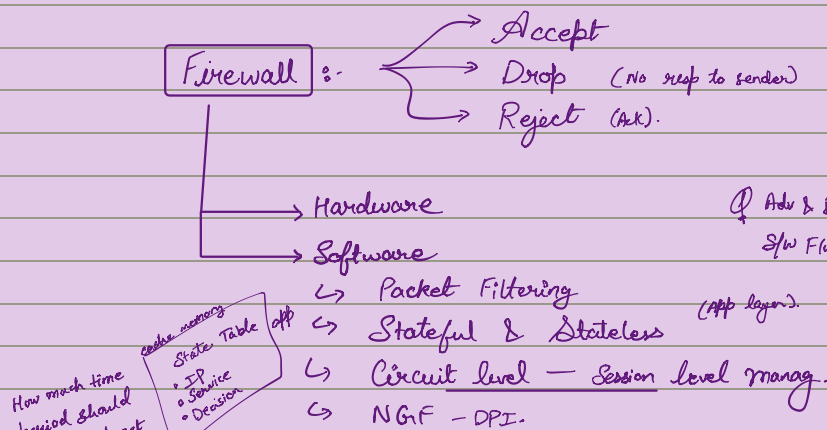
Information Security



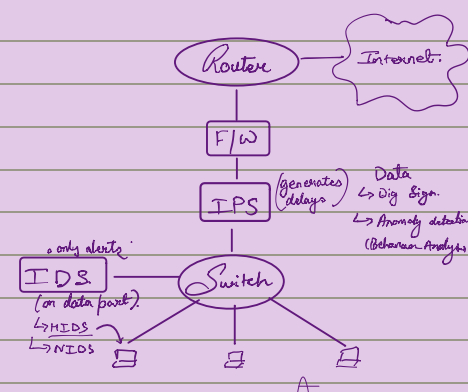
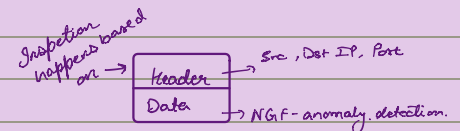
- Hub - No forensic weightage - No memory - Broadcast - Not Secure
- Bridge - LAN Connectors - No memory - Only processing & amplification - No Broadcast - Not Secure - Other Network can see details.
- Switch - multicaster (unic, broad, multi) - MAC & IP - Sniffing, ARP attacks - Forensic Weightage (Identifying IP of the devices from the network at any given time. IP-MAC Binding).
 - CAM Table Flooding → Broadcasting
 - Types :-
 1. Unmanaged
 2. Managed - User friendly dashboard management
 3. LAN
 4. POE - Ethernet & Power available in single line (Comm line)

Q. Working, Layer, Forensic Need, Possible Attacks & Why anyone would attack?

Routers :- 30 to 90 secs - Route Table gets refreshed.



Q. Adv & Disadv of HW & SW FW.



9th Feb.

DNS :- Domain Name System.

↳ Resolves domain name to IP address

↳ 2 ways :- Iterative - provide address to TLD - Local server is contacted everytime
Recursive - TLD will be responsible -
↳ which one is better & why?

In iterative - Local server would get IP address of root, TLD & authoritative
↳ less security - can be attacked.

In recursive - Root Server is centralized - TLD & authoritative would be managed by root

Single & Autonomous
TLD - Diff.

↳ More secure - 13 servers to avoid single point of failure

↳ To prevent overload multiple root servers are provided.

DHCP :- Dynamic Host Configuration Protocol.

↳ Assigning IP address to devices in the network.

yiaddr = ?

Email

↳ User agent - Gmail, outlook,

↓
Browser based,
terminal based,
app based,
messenger based.

SMTP - invoke TCP - establish channel - mail

Server - user agent of receiver & then to user.

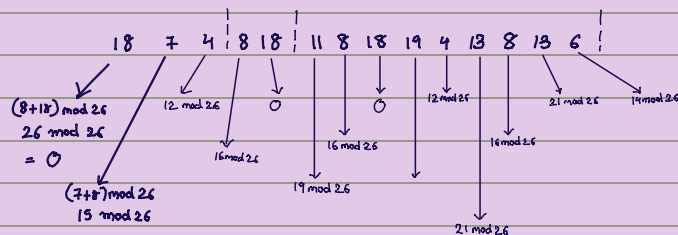
app
(app port, SMTP)
25
code study.
email
(80, 443)

★ Unit - 3. CRYPTOGRAPHY !

IA) Caesar Cipher

K = 8 msg = "She is listening"

Formula = $(P+K) \bmod 26$



HELLO (15)

Decryption

* -15 mod 26

CT = APMQATQABMVQVO

A = 3
Q = 8

m = 2
V = 2

1. Assume E - A = 3

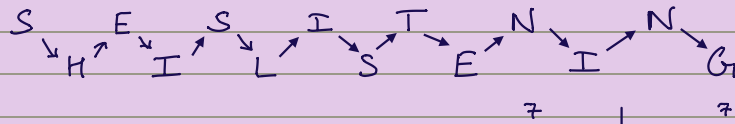
Frequency of letters - occurrence

↳ To remove this problem Polyalphabetic substitution ciphers were developed.

$(P+K) \bmod 26$.

Transposition Cipher.

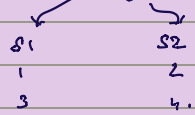
Keyless - Railfence.



$$CT = \text{SESTITNN} | \text{HILSEIG} = 14$$

If its even eg 14 then divide by 2 i.e 7.

SHEISLISTENING.



If odd then extra letter will go to first set.

If multiple tracks are there then divide by that no

6/5/5 if even

5/5/5 if odd.

L T A T C T N G T
E S T A K O I H

$$\frac{17}{2}$$

LTATCTNGT | ESTAKOIH

$$\begin{array}{r} 16+1 \\ 81 \end{array} \quad \begin{array}{r} 16 \\ 82 \end{array}$$

LETSATTACKTONIGHT.

M M T A = 4
E T E T H P R = 7
E A E K = 4

MMTA | ETETHPREK = 15 4, 7, 4.

$$\frac{12}{2} = 6 \quad \frac{11}{2} = 5.5 \quad \frac{27}{2} = 13.5 \quad 7, 13, 6$$

$$\frac{7, 13, 6}{2}$$

$$\frac{27}{2} = 13.5 \quad \frac{13}{2} = 6.5$$

$$7, 13, 7$$

TIYGHCEHSSRPORYETRBICTALV



THIS IS CRYPTOGRAPHY LECTURE.

16th Feb.

A →

Key Transposition.

M M T A L E / E E H R E / E A E K A / T T P P S

25

Key = 4.

M E E T
M E A T
T H E P
A R K P
L E A S
E



5+1

5

if rem=1

5

or

5

5+1

5+1

if rem=2

5
4 2 1
1 2 3 4 5

at first.

5 5 5 5

1 2 3 4 5
13 14 15 16 17

B

1 2 3

S H E I S L I S T E N I N G Z

Padding

Key:

3	1	2
1	2	3

↑ D

Letter Position

E S H L I S T I S I E N Z N G

↓ D

S H E I S L I S T E N I N G Z

C

Repositioning each column.

3 | 5

3 | 5

S	H	E
I	S	L
I	S	T
E	N	I
N	G	Z

→

E	S	H
L	I	S
T	I	S
I	E	N
Z	N	G

→

E L T I Z | S I E N | H S S N G

← D

S-box

P-Box (3 types)

i) Straight

ii) Expansion

iii) Compression.

Substitution > Transposition.

Breaking is complex

DES.

64 bit Plain text

Initial Permutation

→ Reading Key now wise.

Round 1

← K₁

Round 2

← K₂

⋮

Round 16

Final Permutation

Round Key Generator
56-bit

64-bit Key

8 bits are lost as the key from position 8 & 16 is removed.
i.e 4+4.

★ Luck??

* Types of Hashing

- 1] Independent
- 2] Repeated
- 3] Combined
- 4] Sequential
- 5] Hierarchical

* Digital Signature

- Sender uses their private key to 'sign' a digital doc
- Receiver uses 'sender's public key' to verify if sign is valid
- In encryption, receiver's key is used to decrypt the msg whereas here sender's key is used to verify the identity

PRACTICAL

Wireshark Practical

1. ip.addr ==
2. eth for mac add
3. tcp.flags.syn == 1

$$\text{Closed ports} = \text{Total no of ports scanned} - \text{Filtered ports} - \text{Open ports}$$

Statistics

Conversations

Double ports

STATISTICS

1. Conversations
2. Flow
3. Wireshark protocol hierarchy

⇒ When port is open in XMAS: Fin == 0

⇒ Types of scans, XMAS, TCP-SYN, Full Connect, Null, etc (learn)

★ Practical Exam Hack

> frame contains "word"

use all 4 options of mcqs.

RSA

$$p = 13$$

$$q = 17$$

$$\text{public key of A} = 35 = e$$

$$\text{private key of A} = ? (d)$$

$$① \quad m = p \times q = 13 \times 17 = 221$$

$$e = \text{public} \quad d = \text{private}$$

$$② \quad \phi(n) = \phi(221) = (13-1) \times (17-1) = 12 \times 16 = 192$$

$$③ \quad 1 < e < \phi(n) \quad 1 < e < 192$$

co-prime

gcd of e & $\phi(n)$ should be 1.

$$④ \quad de \bmod \phi(n) = 1$$

$$⑤ \quad E \quad CT = PT^e \bmod n$$

$$⑥ \quad D \quad PT = CT^d \bmod n$$

$$\begin{matrix} \text{in} \\ 35 \\ \text{out} \end{matrix} \bmod 221$$

$$d(35) \bmod 192 = 1$$

$$de \equiv 1 \bmod \phi(n)$$

$$de = 1 + K\phi(n)$$

$$d = \frac{1 + K\phi(n)}{e}$$

$$\text{take } K = 0, 1, 2, 3$$

$$\begin{array}{r|l} 0/1 & 2 \\ \hline 1/35 & 1+192 \\ 35/35 & 35 \end{array} \quad \begin{array}{r|l} 1+2 \times 192 & \\ \hline 35 & \end{array}$$

5.5 11
don't take
point

$$e \quad 35, 192$$

$$\begin{array}{r} 5 \overline{) 35} \\ 7 \overline{) 7} \\ 1 \end{array} \quad \begin{array}{r} 2 \overline{) 192} \\ 2 \overline{) 96} \\ 2 \overline{) 48} \\ 2 \overline{) 24} \\ 2 \overline{) 12} \\ 2 \overline{) 6} \\ 3 \overline{) 3} \\ 1 \end{array}$$

23rd Feb.

Unit 5.

Network Forensics.

1. Footprints :- impact that an investigator has on the systems under examination.
Leave less no. of footprints.

2. Evidence :- Any object which proves/disproves an action

- Real :- Physical evidence (HDD, SSD, pendrives, etc.)
- Best :- Directly admissible in the court of law. (photos of crime scene)
- Direct :- Testimony offered by direct witness
- Circumstantial :- doesn't directly support a specific conclusion. It can be linked together with other evidence & used to deduce a conclusion.
- Hearsay :- not a direct witness but heard or seen something, second hand.
- Business Records :- Contracts, agreements, invoices, taxes, logs, msgs
- Digital Ev :- exist in electronic digital form
- Network Based D.E :- comes as a result of communications over a network.

Acquiring volatile data

Acquisition

Content

Storage

Privacy.

Seizure - Operational Continuity

Admissibility -

Bypassing Technique

1. Connect new Router
 2. Re route
 3. Stop main server.
- Small disruption - not overall connectivity.

for switch.

1. Original config has to be done first
2. Unplug main & plug 2nd
3. Few seconds, not major.

Network Forensics Investigative Methodology

Incident :-

O - Obtain Info : What happened, process, ppl involved, actions taken after discovery, Summary, Incident manager & process, Legal issues, Time frame

Environment: Business model, Network Topology, Available sources of evidence, Org Structure, IRM process, Comm System, Overall Resources available.

S - Strategize : List resources, Prioritize evidence acquisition, Plan initial acquisition

C - Collect evidence:

A - Analyse:

R - Report:

Prioritizing

1. Volatile data
2. Storage seg vs storage you have.
- 3.

Questions

1. Diff b/w capture filter & display filter.
2. Nmap as root & non root user permissions. What are the differences

[1] Caesar Cipher (Additive, Shift)

↳ aka monoalphabetic

Eg 1; PT = HELLO K=3

Formula = $(P+K) \bmod 26$

Encryption

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26

CT = KHOOR

↳ To decrypt do minus K.

Formula = $(C-K) \bmod 26$

Decryption

Q What to do if you get negative mod?

Frequency : ETAOINSHRDLG...

[2] Polyalphabetic Ciphers

↳ Each occurrence of a character may have a diff substitute. One-to-many. More secure

VIGENERE CIPHER (By: Blaise de Vigenere). French mathematician

1850s

$$P_i = (C_i - K_i) \bmod 26$$

Key	S	T	O	P	S	T	O	P	S	T	O	P	S
Text	T	O	D	A	Y	I	S	T	H	E	D	A	Y
Text letters	19	14	3	0	24	8	18	19	7	4	3	0	24
Key letters	18	19	14	15	18	19	14	15	18	19	14	15	18
TextL+Key L	37	33	17	15	42	27	32	34	25	23	17	15	42
mod26 ans	11	7	17	15	16	1	6	8	25	23	17	15	16
	L	H	R	P	Q	B	G	I	Z	X	R	P	Q

LHRP

$(12-18) \bmod 26$
 $-6 \bmod 26$
6

If there is a -ve no then add mod K bad vala no before.

[3] Transposition Ciphers

★ If Key Transposition is there

E	3	1	2	4	5
D	1	2	5	4	3

Divide sentence in these parts & add 'z' or any bogus character at end to make same size group

↳ Above one was letter by letter, we can even do column by column

★ Keyless Ciphers

↳ Railfence

I M A T A
A A S H

CT = IMATAAASH

K=2

$\frac{9}{2} = 5, 4$

↳ Column 20 now & vice versa K=5

I A M A A CT = I S I R G | A T S F I | M H S E R | A A P C L | A M E T
S T n A M
I S S P E
R F E C T
G I R L

Read like →

★ Rotation cipher is rotate the whole letter by the number of Key given

$$5 \overline{) 24} \begin{matrix} 4 \\ 20 \\ 4 \end{matrix}$$

$$\begin{matrix} 4+1 \\ 4+1 \\ 4+1 \\ 4+1 \\ 4 \\ \hline 24 \end{matrix}$$

Key
• letter

Keyless
• Railfence
• C & R

Comb

• Row + Box Key